2.0 Contraintes et déformations

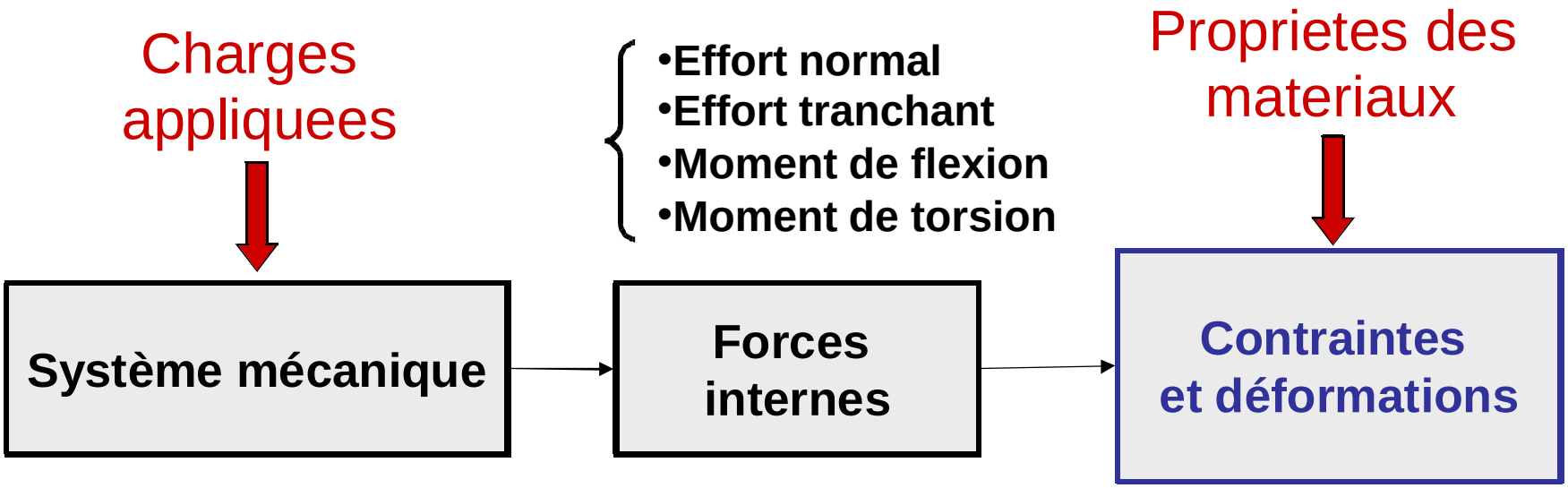


Plan

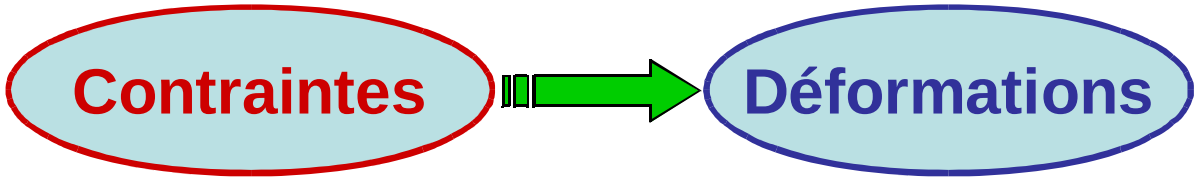
1. Introduction
2. Contrainte normale
3. Déformation normale
4. Contrainte de cisaillement
5. Déformation de cisaillement
6. Contrainte admissible
7. Conclusion
8. Problèmes

Hibbeler: Sections 1.3, 1.4, 1.5, 1.6, 1.7, 2.1, 2.2

1. Introduction



- Contrainte: force par unite de surface qui agit a l'interieur d'un corps (intensite des forces internes)
- Deformation: mesure du changement de forme du corps



2. Contraintes et forces

$$\sigma_z = \lim_{\Delta A \to 0} \frac{\Delta F_z}{\Delta A}$$

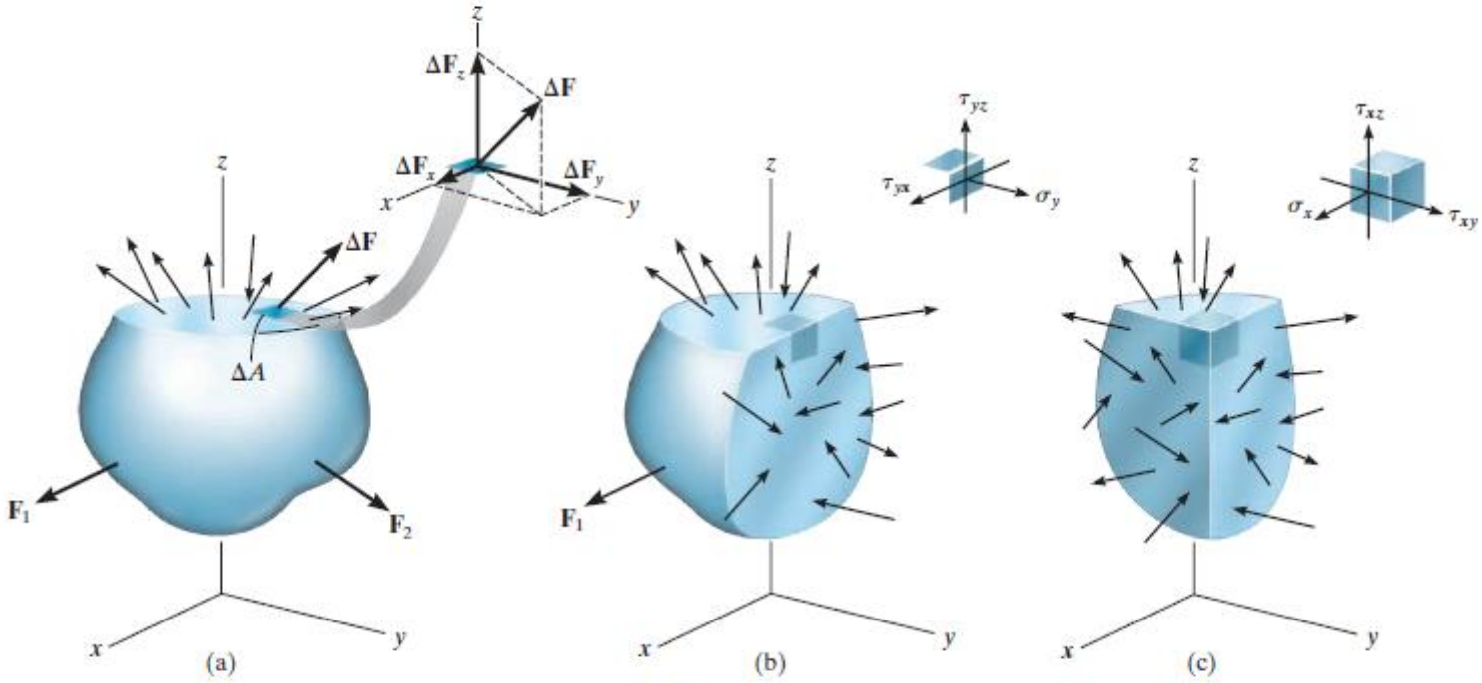


Fig. 1–10

2. Contraintes et forces

$$\sigma_z = \lim_{\Delta A \to 0} \frac{\Delta F_z}{\Delta A}$$

$$\tau_{zx} = \lim_{\Delta A \to 0} \frac{\Delta F_x}{\Delta A}$$

$$\tau_{zy} = \lim_{\Delta A \to 0} \frac{\Delta F_y}{\Delta A}$$

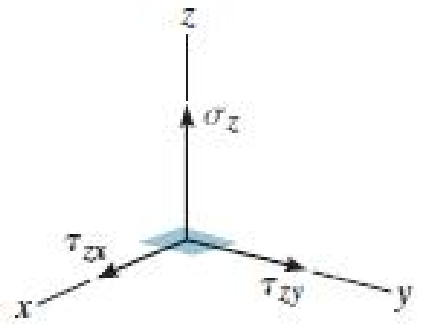


Fig. 1–11

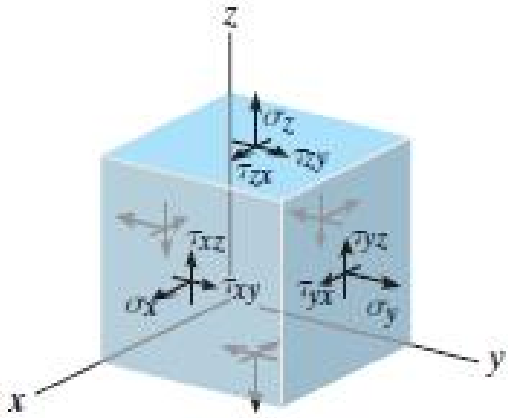
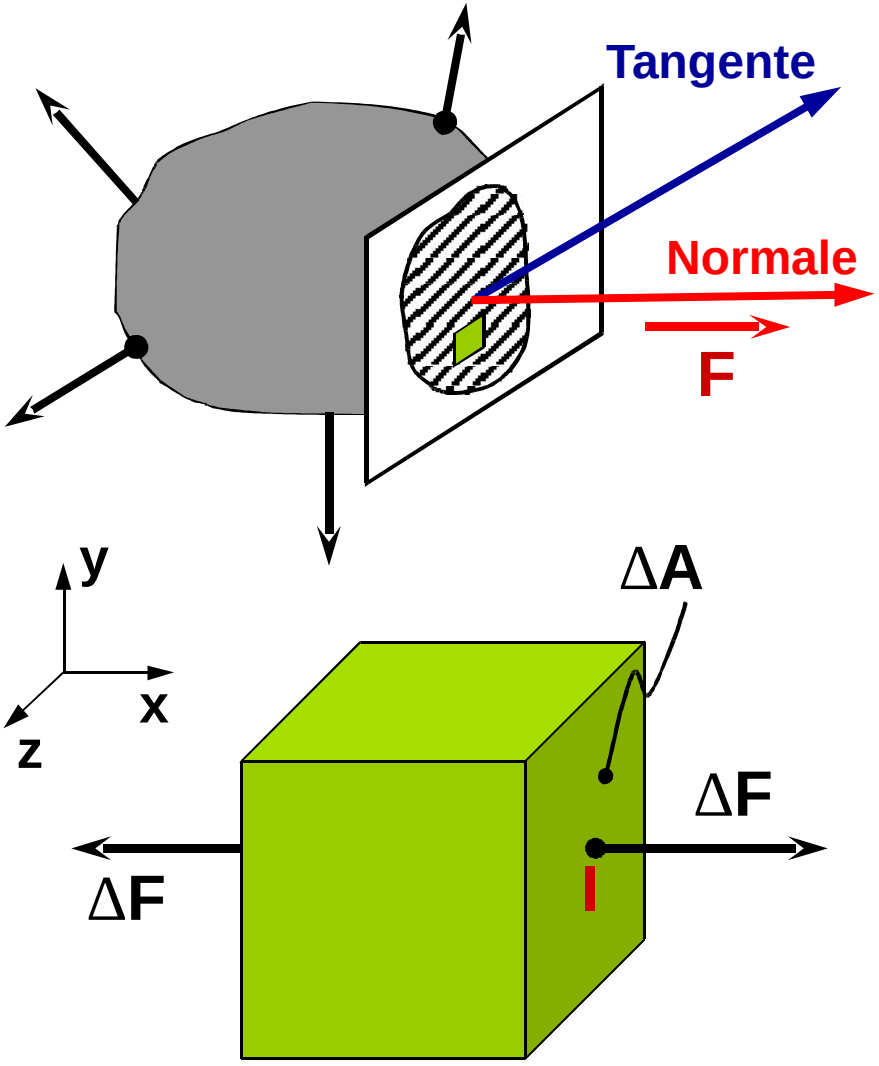


Fig. 1–12

2. Contrainte normale



2.1 Définition:

La **contrainte normale** σ est générée par l'effort interne qui agit dans la direction **normale** à la surface considérée.

Soit une force interne qui agit dans la direction x.La contrainte normale au point I dans la direction x, σ ,est définie par:

$$\sigma = \lim_{\Delta A \to 0} \frac{\Delta F}{\Delta A}$$

Unité SI: N/m² ou Pa "Pascals"

Unité FPS : lb/in² ou Psi

2. Contrainte normale

Unités SI pour les contraintes :

Pa (Pascals) : **1 Pa = 1 N/m²** c'est l'unité "officielle"

Cependant l'unité la plus pratique et la plus couramment utilisée :

MPa (MégaPascals) : **1 MPa = 10^6 Pa = 1 N/mm²**

Ce qui permet de travailler toutes les dimensions en mm.

On utilisera aussi le **GPa = 1 kN/mm² = 1000 MPa**

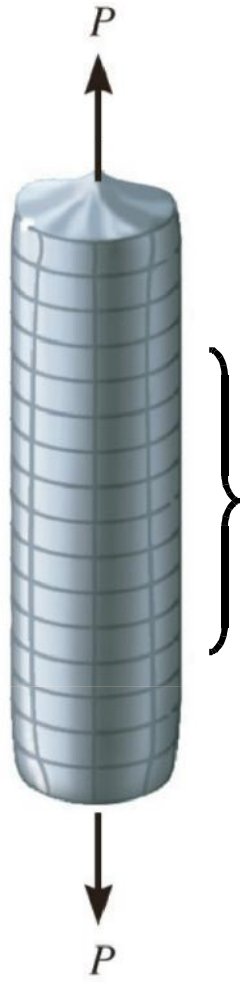
Dans le système américain :

psi = 1 livre/pouce² ou **lb/in²** unité officielle

ksi = 1000 psi

Conversion : 1 MPa = 145 psi

2.2 Contrainte normale moyenne dans une barre



Déformation uniforme

Soit une barre soumise à une force axiale P.

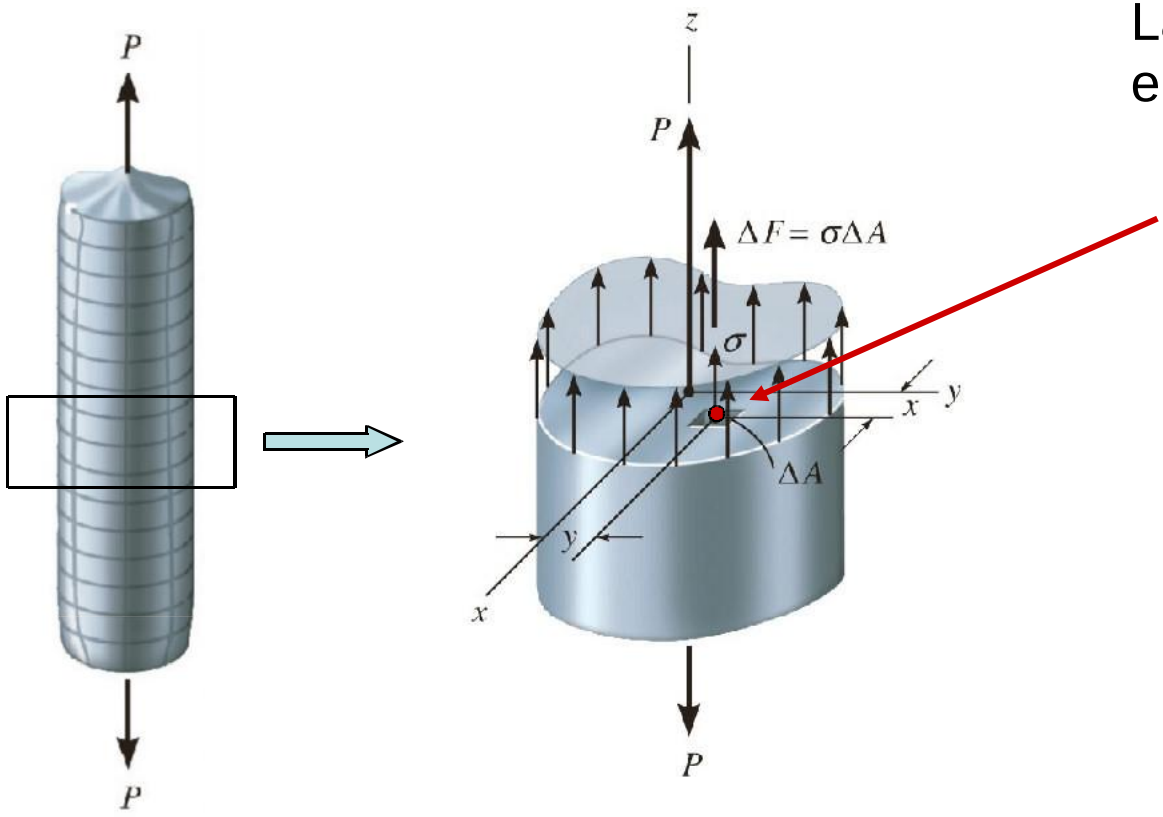
Hypothèses:

1) Barre prismatique
2) Barre homogène

→ Déformation uniforme

→ Contrainte uniforme

2.2 Contrainte normale moyenne dans une barre



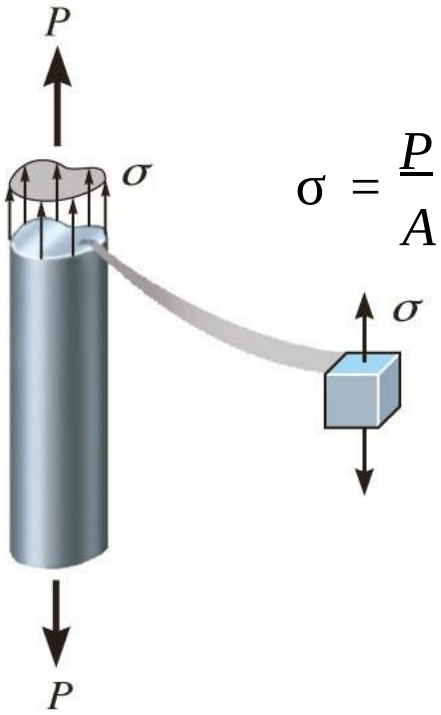
La contrainte normale en un point:

$$\sigma = \lim_{\Delta A \to 0} \frac{\Delta F}{\Delta A}$$

La contrainte normale moyenne:

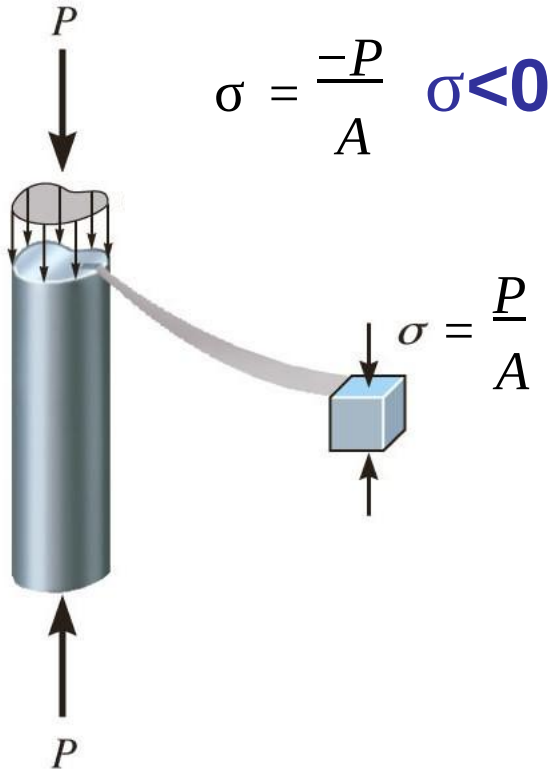
$$\sigma_{moy} = \frac{P}{A}$$

2.3 Convention de signe:



$$\sigma = \frac{P}{A}$$ **σ>0**

Tension



$$\sigma = \frac{-P}{A}$$ **σ<0**

$$\sigma = \frac{P}{A}$$

Compression

2. Contrainte normale

Procedure for Analysis

The equation $\sigma = P/A$ gives the *average* normal stress on the cross-sectional area of a member when the section is subjected to an internal resultant normal force **P**. For axially loaded members, application of this equation requires the following steps.

Internal Loading.

- Section the member *perpendicular* to its longitudinal axis at the point where the normal stress is to be determined and use the necessary free-body diagram and force equation of equilibrium to obtain the *internal axial force* **P** at the section.

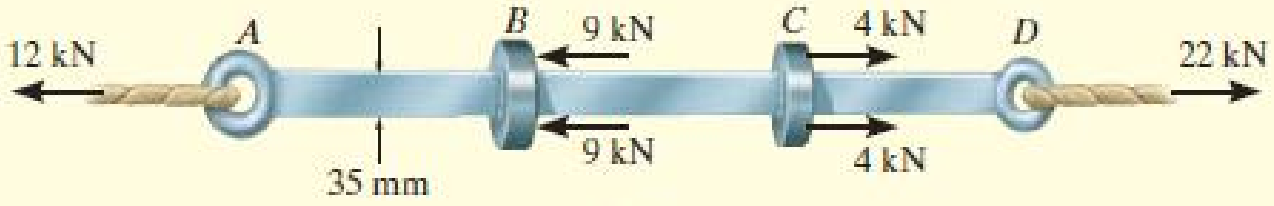
Average Normal Stress.

- Determine the member's cross-sectional area at the section and calculate the average normal stress $\sigma = P/A$.
- It is suggested that σ be shown acting on a small volume element of the material located at a point on the section where stress is calculated. To do this, first draw σ on the face of the element coincident with the sectioned area A. Here σ acts in the *same direction* as the internal force **P** since all the normal stresses on the cross section develop this resultant. The normal stress σ on the other face of the element acts in the opposite direction.

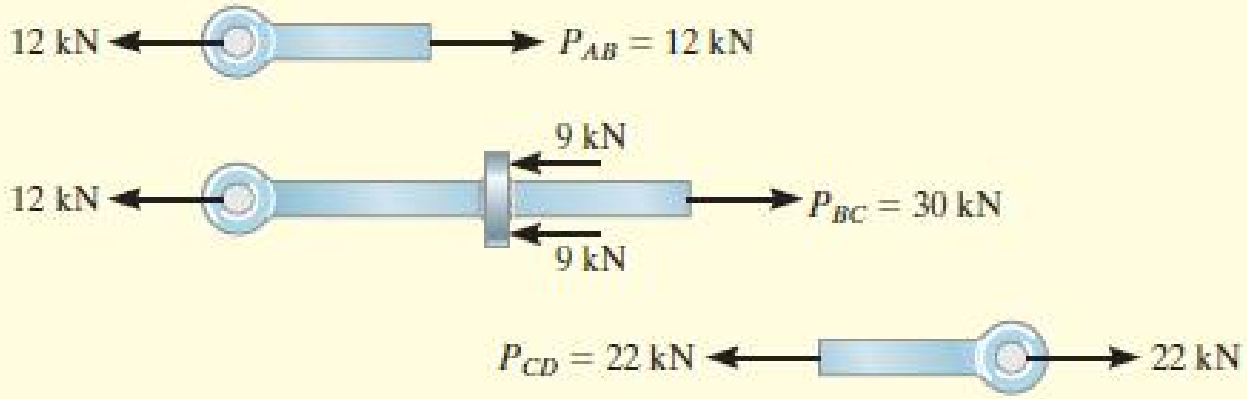
2. Contrainte normale

Exemple_1

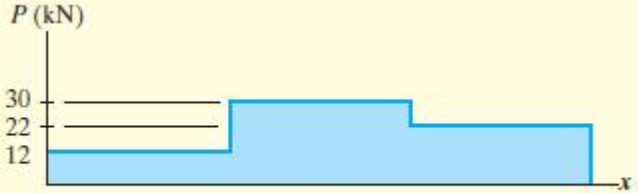
The bar in Fig. 1–16*a* has a constant width of 35 mm and a thickness of 10 mm. Determine the maximum average normal stress in the bar when it is subjected to the loading shown.



(a)



(b)

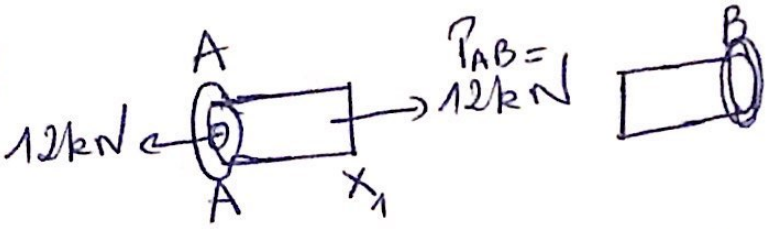


(c)

Exemple 1 :

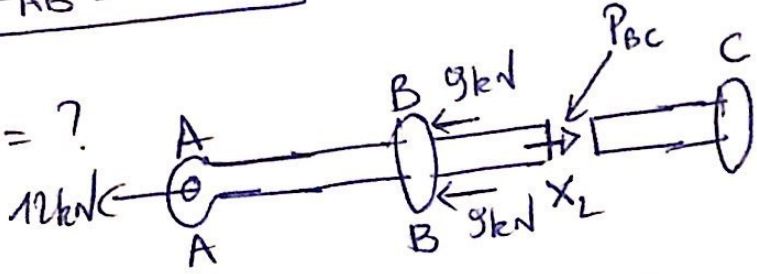
On sait que $\sigma = \frac{P}{A} \Rightarrow \sigma_{max} = \frac{P_{max}}{A}$; ici $A = cte$

$P_{AB} = ?$



$\Rightarrow \boxed{P_{AB} = 12kN}$ car le morceau AX_1 est en équilibre

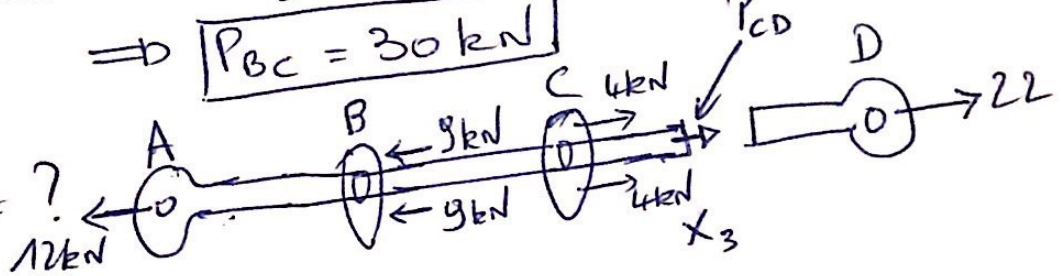
$P_{BC} = ?$



Le morceau AX_2 est en équilibre donc $12 + 9 + 9 - P_{BC} = 0 (kN)$

$\Rightarrow \boxed{P_{BC} = 30kN}$

$P_{CD} = ?$

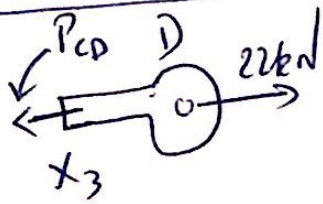


Le morceau AX_3 est en équilibre donc :

$12 + 9 + 9 - 4 - 4 - P_{CD} = 0 (kN)$

$\Rightarrow \boxed{P_{CD} = 22kN}$

ou bien :



le morceau X_3D est en équilibre

Donc $P_{CD} - 22 = 0 \Rightarrow \boxed{P_{CD} = 2kN}$

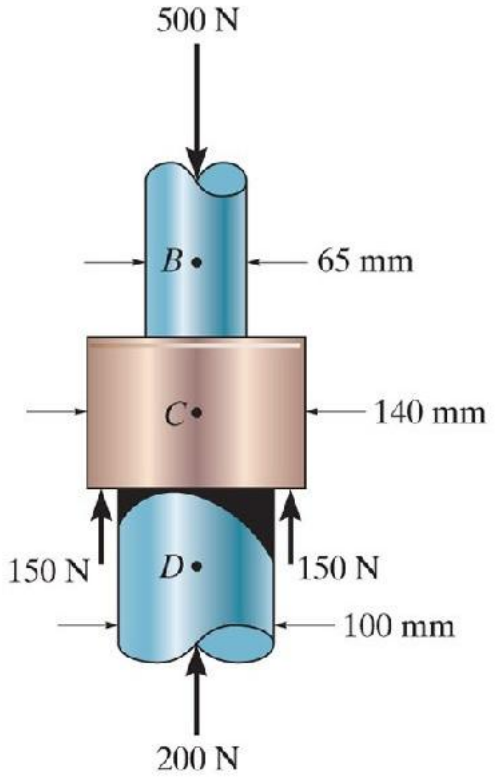
La section $A = 35 \times 10^{-3} \times 10.10^{-3} = 350.10^{-6} m^2 = cte$

$A = cte \Rightarrow \sigma_{max} = \frac{P_{BC}}{A} = \frac{30.10^3 N}{350.10^{-6} m^2} = 85,71 MPa$

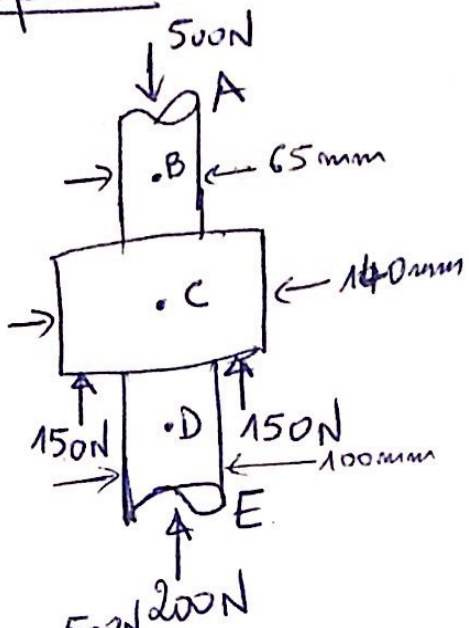
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Exemple 2

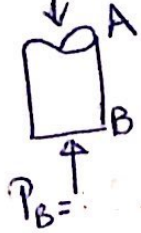
Un palier de butée est soumis au chargement ci-contre. Déterminer la contrainte normale moyenne aux sections passant par les points B, C et D. Les représenter sur un élément de volume.



Exemple 2 :



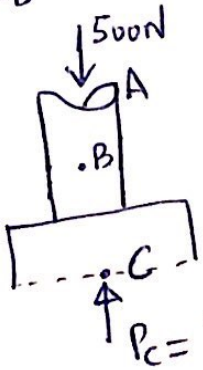
$P_B = ?$



$\Rightarrow$ $P_B - 500N = 0 \Rightarrow P_B = 500N \Rightarrow \sigma_B = \frac{500}{\left(\frac{65.10^{-3}}{2}\right)^2 \pi}$

Car le morceau AB est en équilibre $\Rightarrow \sigma_B = 150679,2 P_a$

$P_C = ?$

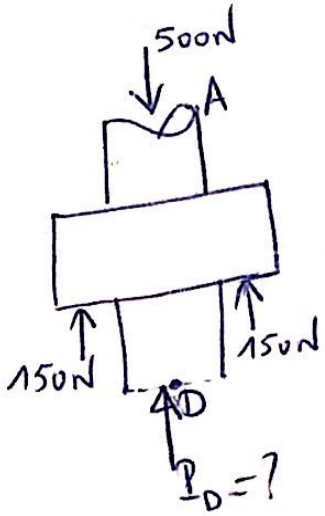


$P_c - 500N = 0 \Rightarrow P_c = 500N$

Morceau AC en équilibre $\Rightarrow \sigma_c = \frac{500}{\left(\frac{140.10^{-3}}{2}\right)^2 \pi}$

$= 32480,6 P_a$

$P_D = ?$



Le morceau AD en équilibre donc

$-500 + 150 + 150 + P_D = 0$

$\Rightarrow P_D = 200 N$

$\Rightarrow \sigma_D = \frac{200}{\left(\frac{100.10^{-3}}{2}\right)^2 \pi} = 25464,8 P_a$

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2. Contrainte normale

Exemple 3

Member AC shown in Fig. 1–19*a* is subjected to a vertical force of 3 kN. Determine the position x of this force so that the average compressive stress at the smooth support C is equal to the average tensile stress in the tie rod AB. The rod has a cross-sectional area of 400 mm^2 and the contact area at C is 650 mm^2.

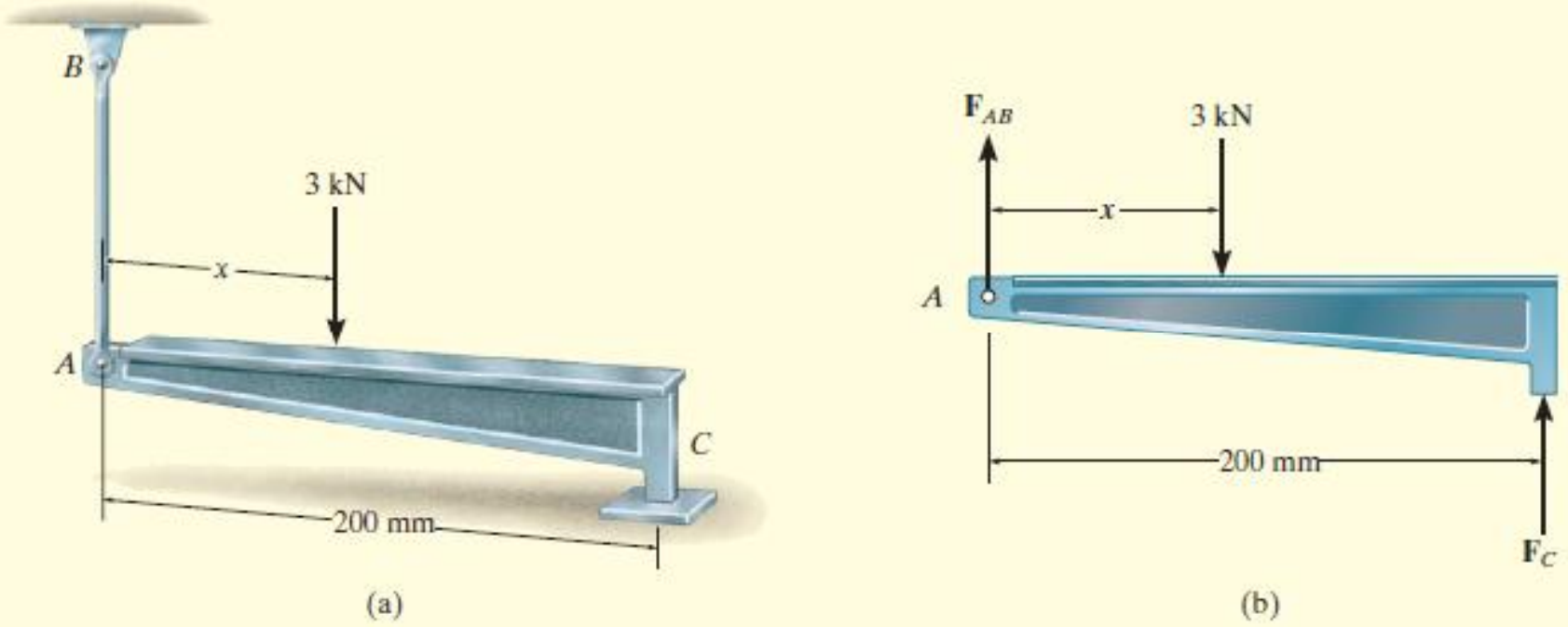
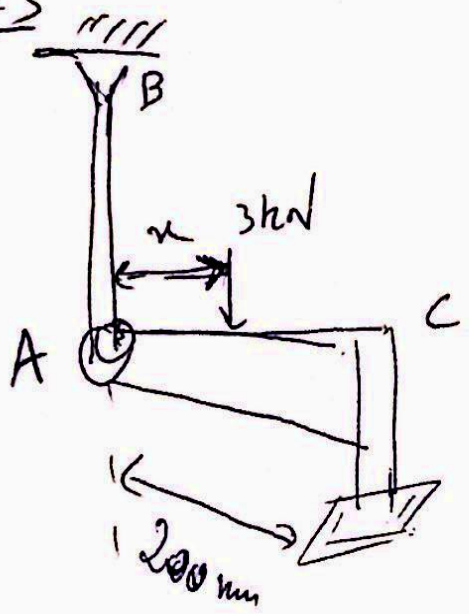


Fig. 1–19

Exple 3

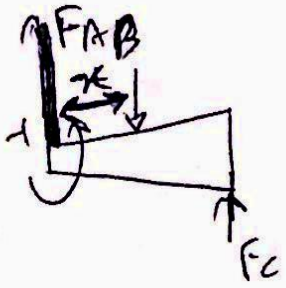


$x = ?$ tg : $\sigma_C = \sigma_{AB}$

$$\sigma_{AB} = \frac{F_{AB}}{A_{AB}} \; ; \; \sigma_C = \frac{F_C}{A_C}$$

Equilibre $\Rightarrow F_{AB} + F_C - 3kN = 0$

$$\sum M_A = 0 \Rightarrow -3000x + (200 \quad) F_C = 0$$



$$\begin{cases} F_{AB} + F_C - 3000 = 0 \\ -3000x + 200\,mm\, F_C = 0 \end{cases}$$

$$\text{or } \frac{F_{AB}}{A_{AB}} = \frac{F_C}{A_C} \Rightarrow F_{AB} = F_C \times \frac{A_{AB}}{A_C}$$

$A_{AB} = 400 mm^2$

$A_C = 650 mm^2$ $\Rightarrow F_{AB} = F_C \times \frac{1}{1,62}$

$$F_{AB} + F_{AB} \times 1,62 - 3000 = 0$$

$$2,62 F_{AB} = 3000 \Rightarrow F_{AB} = 1142,8\,?$$

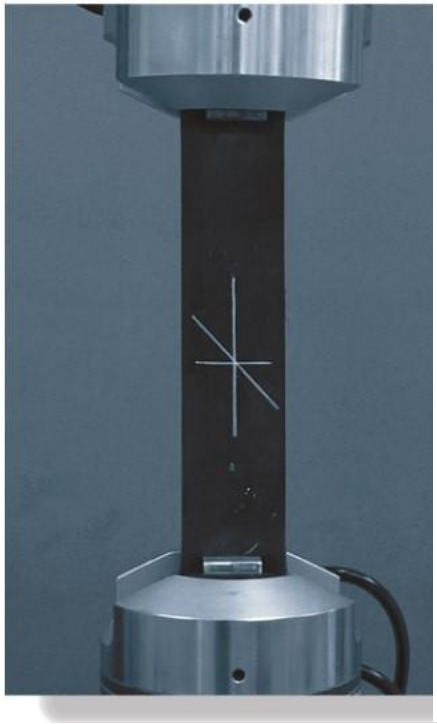
$$F_C = 1857,14$$

$$x = \frac{F_C \times 200}{3000} = 123,8\,mm$$

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3. Deformation normale

Éprouvette en traction



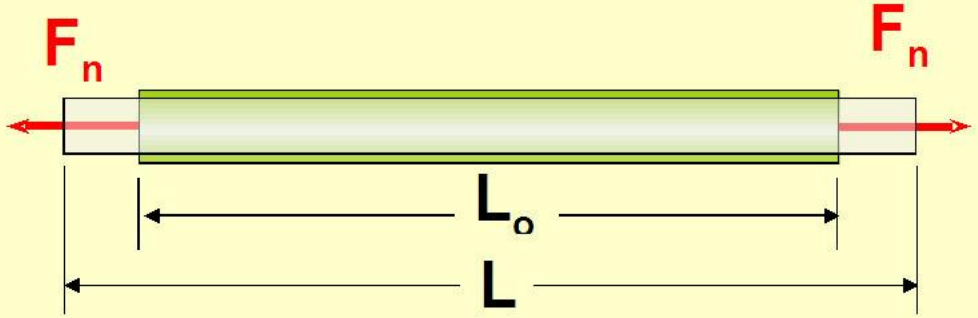


allongement

3. Deformation normale

3.1 Définition:

Déformation normale= $\dfrac{\text{Changement de Longueur}}{\text{Longueur initiale}}$

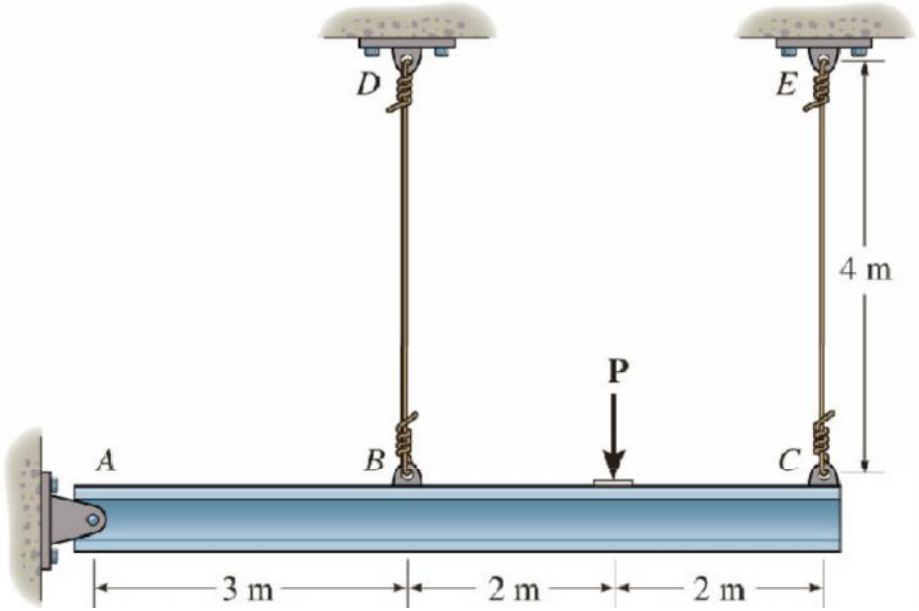


$$\varepsilon = \frac{\Delta L}{L_o} = \frac{L - L_o}{L_o}$$

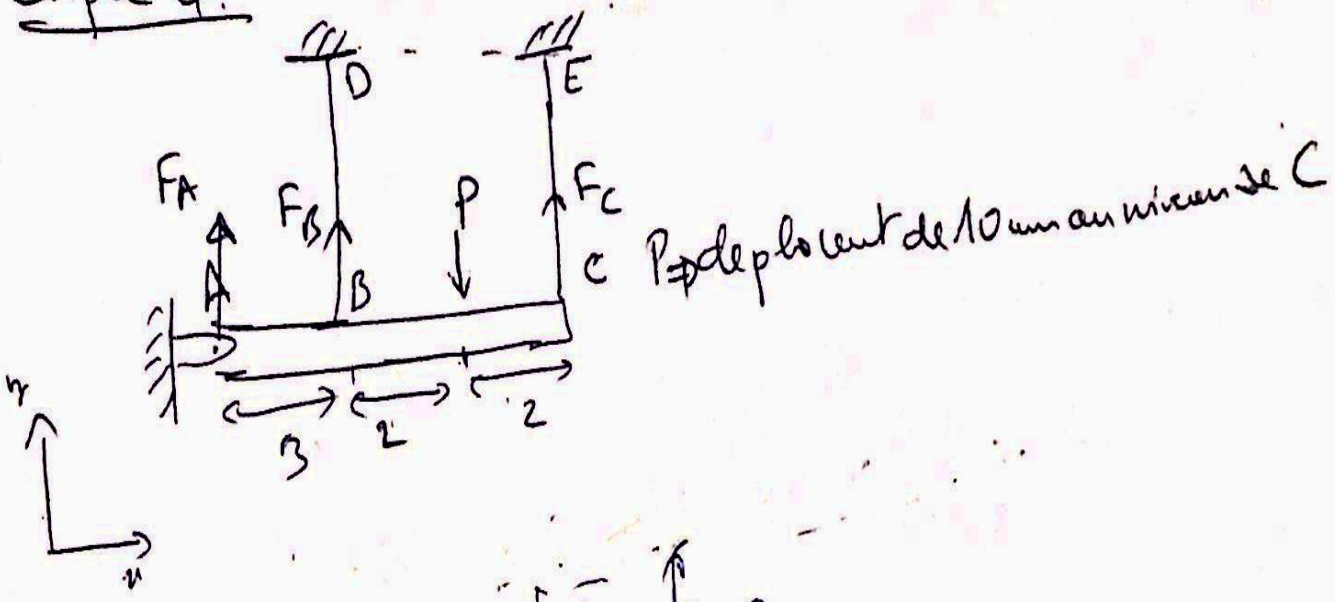
Unité: adimensionnel m/m, μm/m, με, in/in

3.2 Exemple 4:

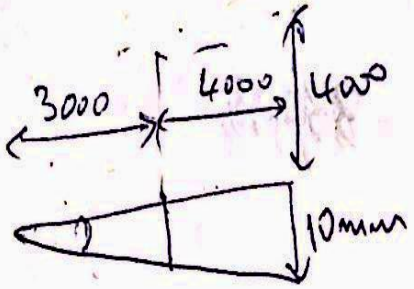
Soit une barre rigide soutenue par un pivot au point A et deux cordes aux points B et C. Si la charge P induit au point C un déplacement vers le bas de 10 mm, calculer les déformations normales dans les cordes BD et CE.



Exple 4:



P déplacement de 10 mm au niveau de C



$$\varepsilon_C = \frac{(4000+10)-4000}{4000} = 0,0025 \quad , \quad \Delta C = 10\,mm.$$

$\varepsilon_D = ?$ $\Delta_B = ?$ $tg\alpha = \frac{10}{7000} = \frac{\Delta B}{3000} \Rightarrow \Delta B = \frac{3000 \times 10}{7000}$

$$\Delta B = 4,287\,mm$$

$$\varepsilon_B = \frac{(4000+4,287)-4000}{4000} = 0,00107$$

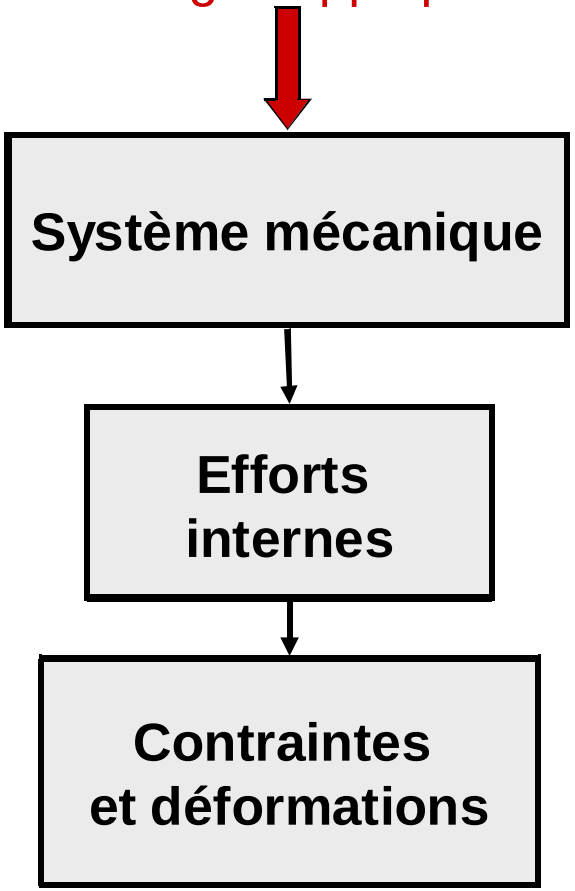
12

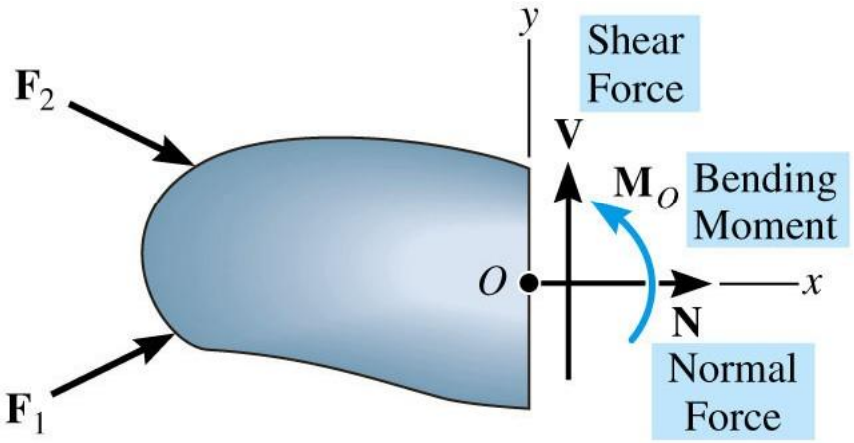
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4. Contrainte de cisaillement

4.1 Rappel:

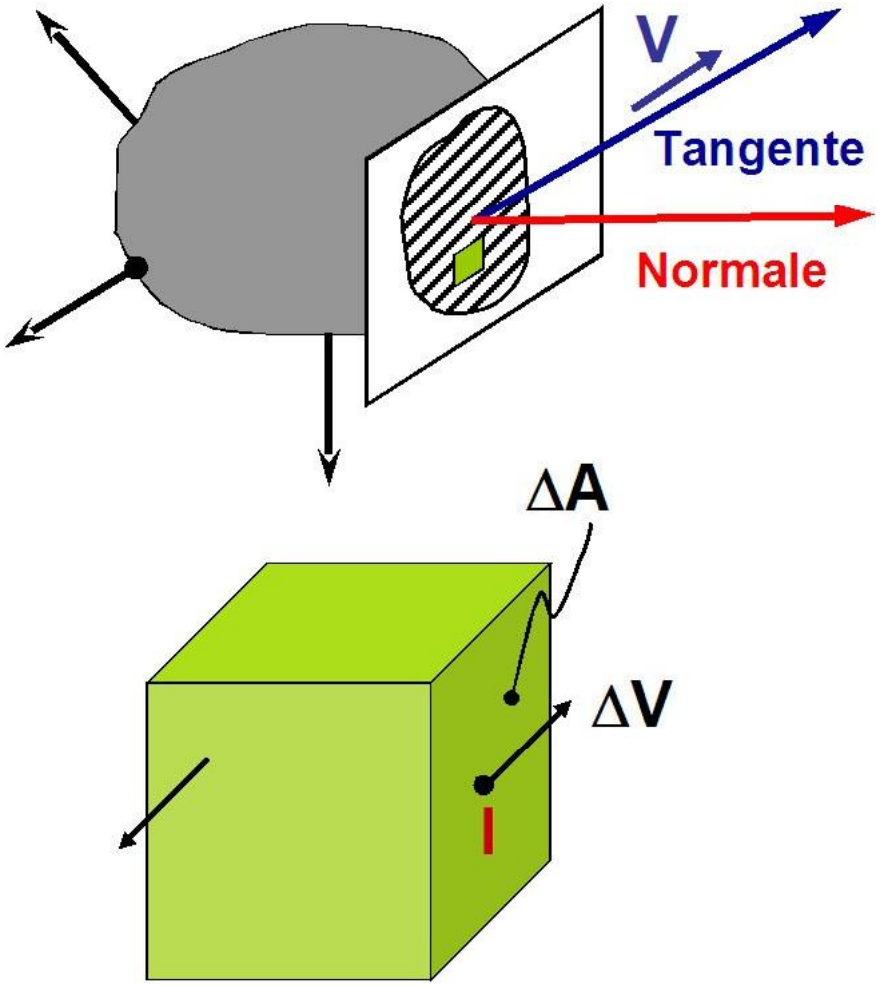
Charges appliquées





- **Effort normal**
- **Effort tranchant**
- **Moment de flexion**
- **Moment de torsion**

4. Contrainte de cisaillement

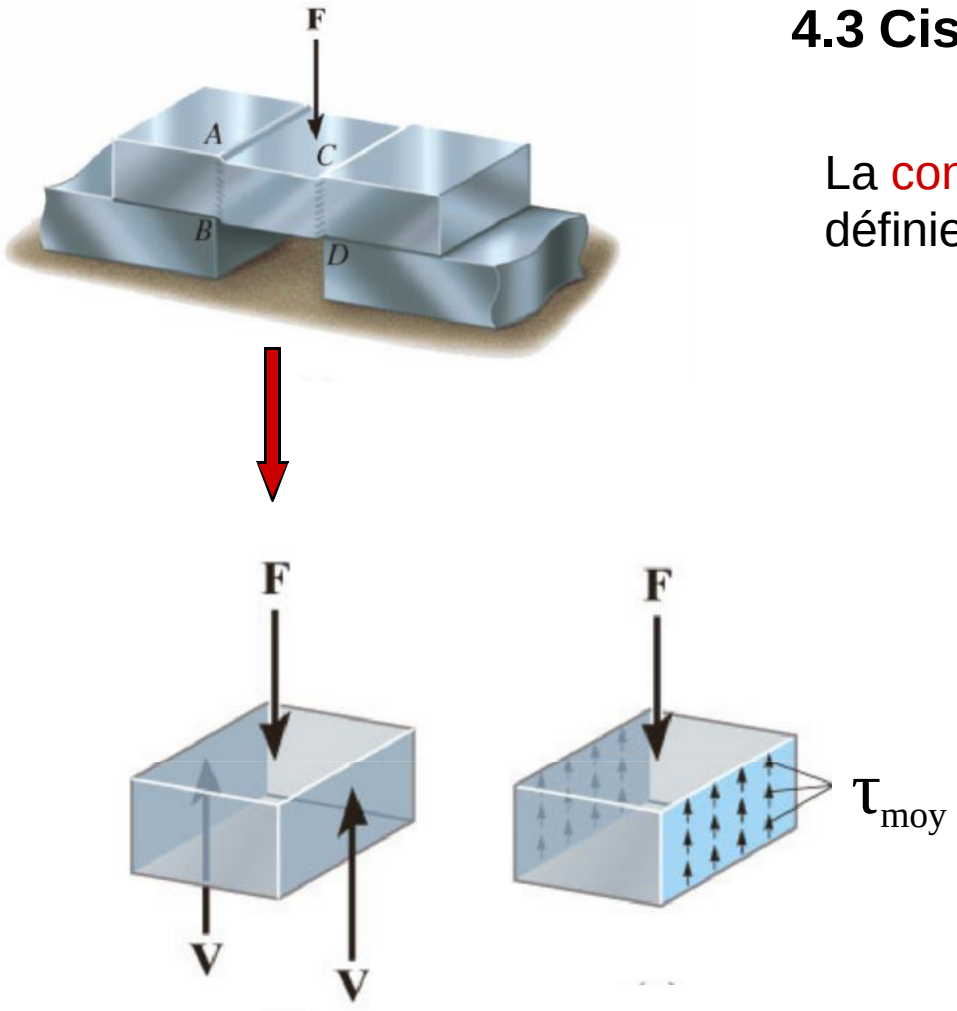


4.2 Définition:

La contrainte de cisaillement est générée par la force qui agit parallèlement à la surface interne considérée (tangente).
La contrainte de cisaillement au point I, τ ,est définie par:

$$\tau = \lim_{\Delta A \to 0} \frac{\Delta V}{\Delta A}$$

4. Contrainte de cisaillement



τ_{moy}

4.3 Cisaillement moyen:

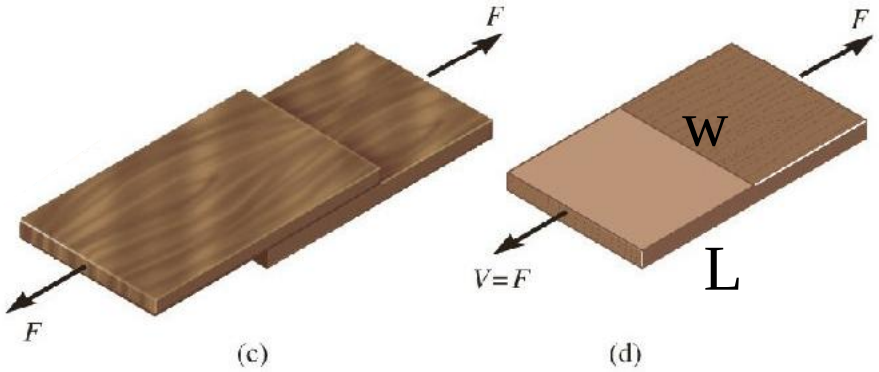
La contrainte de cisaillement moyenne est définie par:

$$\tau_{moy} = \frac{V}{A}$$

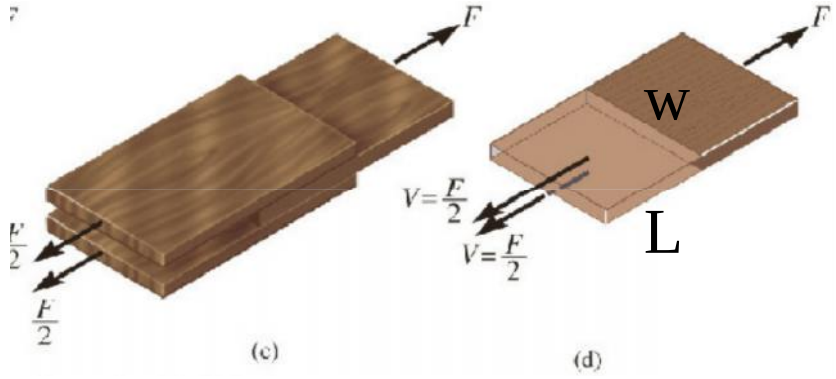
Unité SI: N/m² ou Pa

Unité FPS : lb/in² ou Psi

4.4 Cisaillement simple et cisaillement double:



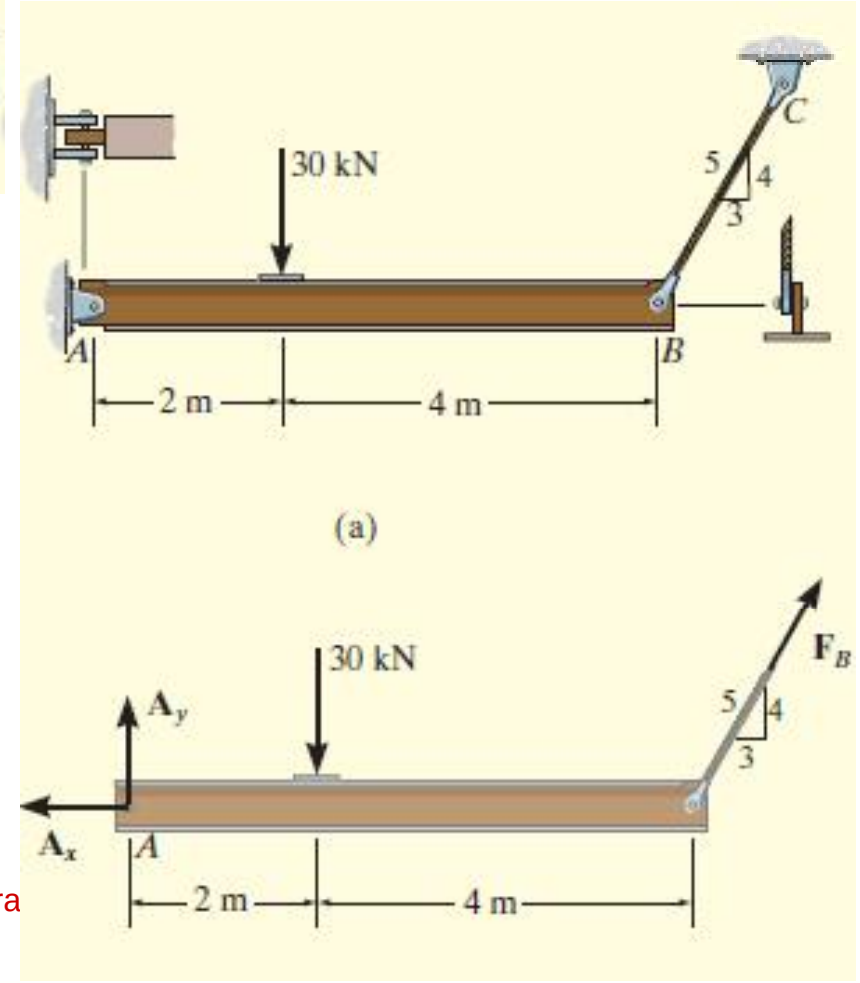
1 plan de cisaillement



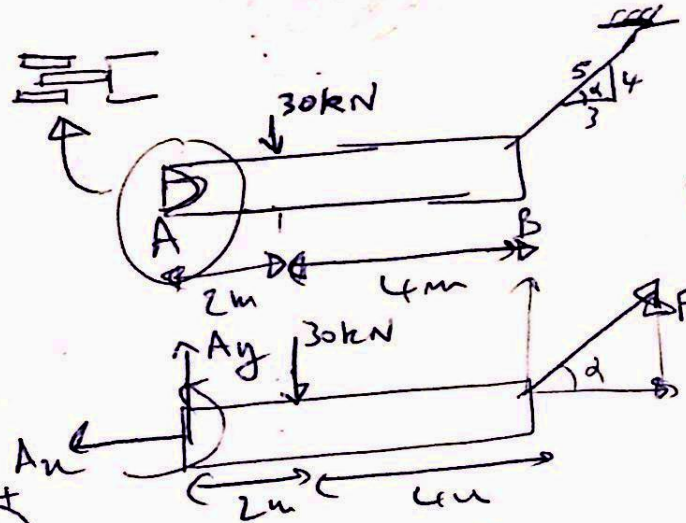
2 plans de cisaillement

4.5 Exemple 5

Determine the average shear stress in the 20-mm-diameter pin at *A* and the 30-mm-diameter pin at *B* that support the beam in Fig. 1–22*a*.



Exples :



$\tau_A = ?$ $\tau_B = ?$

$\cos\alpha = F_{Bx}/F_B = \frac{3}{5} \Rightarrow F_{Bx} = F_B \frac{3}{5}$

$\sin\alpha = F_{By}/F_B = \frac{4}{5} \Rightarrow F_{By} = \frac{4}{5} F_B$



$\Sigma F_x = 0 \Rightarrow -A_x + F_{Bx} = 0 \Rightarrow A_x = F_{Bx} = F_B \times \frac{3}{5}$

$\Sigma F_y = 0 \quad A_y - 30 + F_{By} = 0 \Rightarrow A_y = -F_{By} + 30 = F_B \times \frac{4}{5} + 30$

$\Sigma M_A = 0 \Rightarrow -2 \times 30 + 6 \times F_{By} = 0 \Rightarrow +60 = 6 F_{By} \Rightarrow F_{By} = 10$

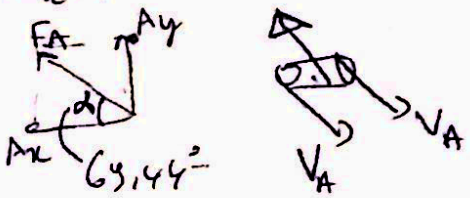
$\Rightarrow F_B = \frac{5}{4} F_{By} = \frac{50}{4} = 12{,}5\ kN \Rightarrow A_y = 30 - F_{By}$

$= 20\ kN$

$A_x = F_B \times \frac{3}{5} = \frac{5}{4} \times \frac{3}{5} \times F_{By} = \frac{3}{4} F_{By} = \frac{30}{4} = 7{,}5\ kN$

$F_A = \sqrt{(7{,}5)^2 + 20^2} = 21{,}36\ kN$

$\tan\alpha_A = \frac{A_y}{A_x} \Rightarrow \alpha = 69{,}44°$



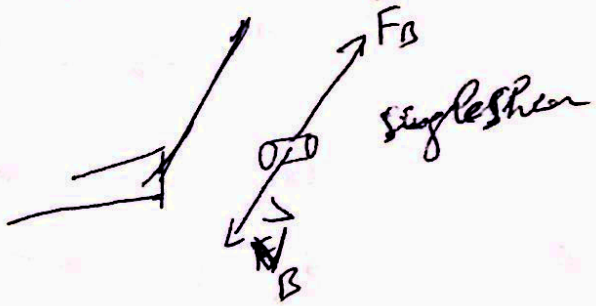
double cisaillement

$V_A = \frac{F_A}{2} = 10{,}68\ kN$

$\tau_A = \frac{V_A}{A_n} = \frac{10{,}68\ kN}{\left(\frac{20.10^{-3}}{2}\right)^2 \pi} = 33{,}99\ MPa \simeq 34\ MPa$

$F_B = 12{,}5\ kN$

$\tau_B = \frac{F_B}{A_B} = \frac{12{,}5.10^3\ N}{\left(\frac{30.10^{-3}}{2}\right)^2 \pi} = 17683.\ Pa$

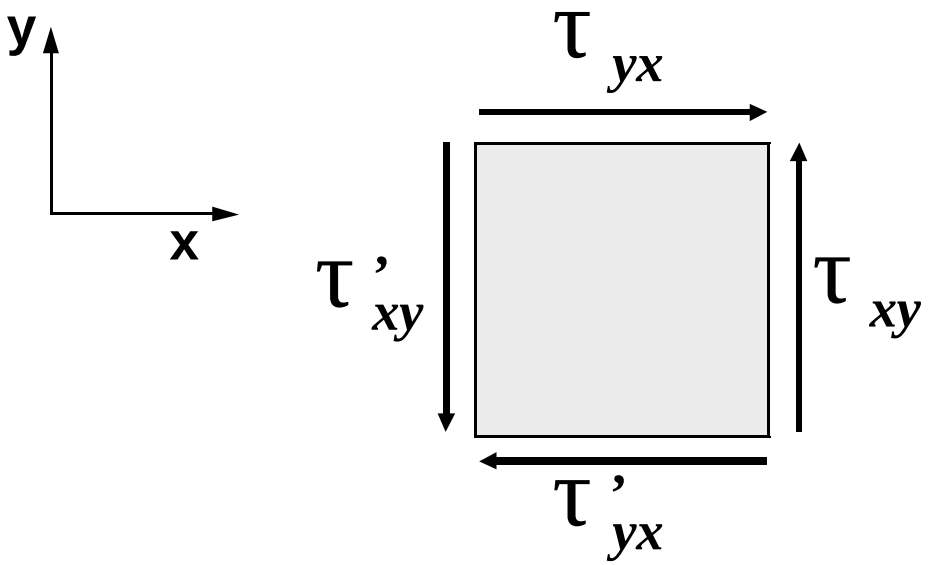


Voir page 12 du livre

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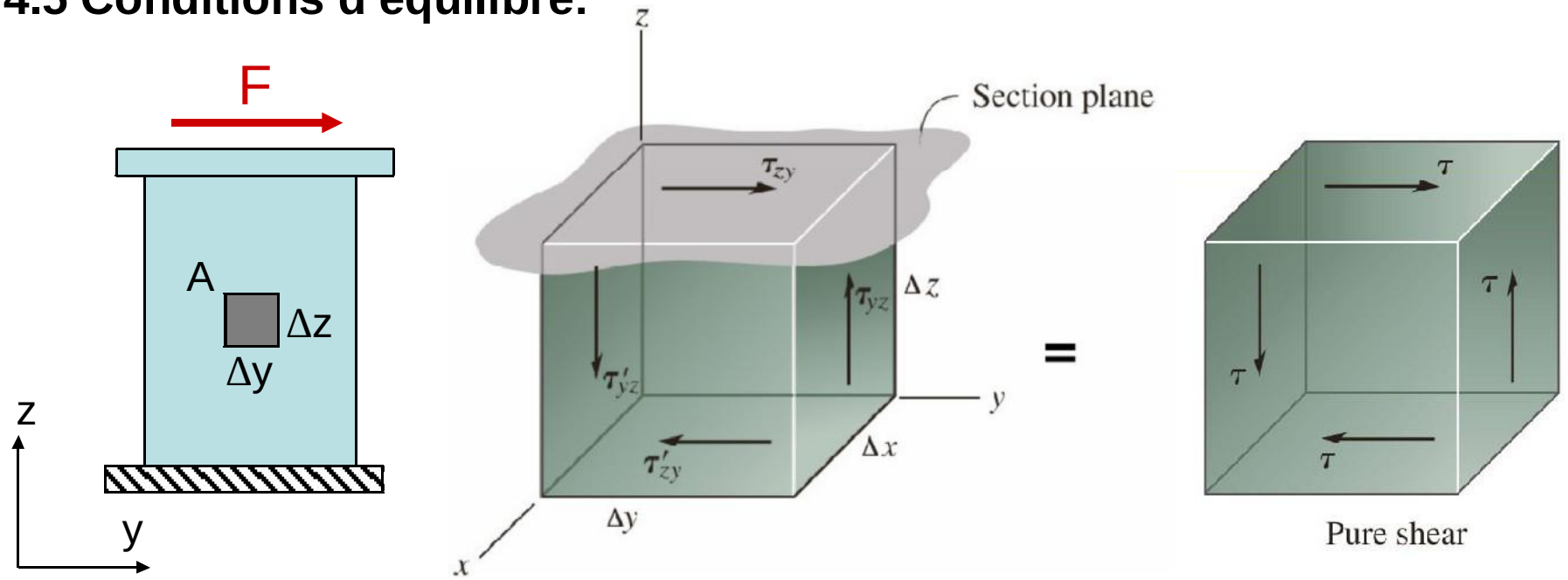
4.4 Notations:

La contrainte de cisaillement est identifiée à l'aide de deux indices. Le premier indice indique la direction de la normale à la surface sur laquelle la contrainte agit. Le second indice indique la direction de la force interne.



Relation entre τ_{xy} , τ_{yx}, τ'_{xy} , τ'_{yx}? (équilibre)

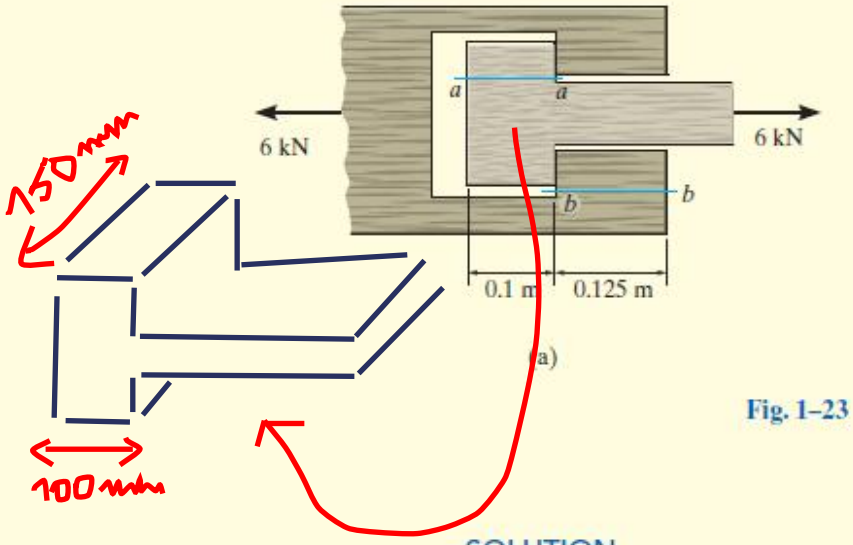
4.5 Conditions d'équilibre:



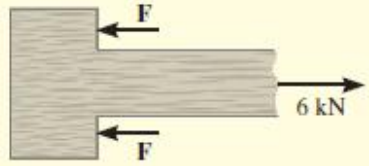
$$F_y = 0 \qquad F_z = 0 \qquad M_x = 0$$

Exemple 6

If the wood joint in Fig. 1–23*a* has a width of 150 mm, determine the average shear stress developed along shear planes *a*–*a* and *b*–*b*. For each plane, represent the state of stress on an element of the material.



(a)



(b)

Fig. 1–23

SOLUTION

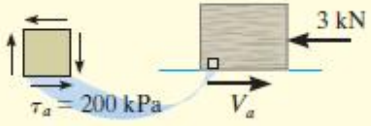
Internal Loadings. Referring to the free-body diagram of the member, Fig. 1–23*b*,

$$\stackrel{+}{\rightarrow} \Sigma F_x = 0; \qquad 6\ \text{kN} - F - F = 0 \qquad F = 3\ \text{kN}$$

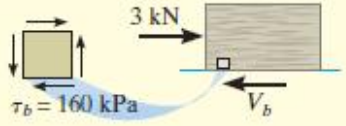
Now consider the equilibrium of segments cut across shear planes *a*–*a* and *b*–*b*, shown in Figs. 1–23*c* and 1–23*d*.

$$\stackrel{+}{\rightarrow} \Sigma F_x = 0; \qquad V_a - 3\ \text{kN} = 0 \qquad V_a = 3\ \text{kN}$$

$$\stackrel{+}{\rightarrow} \Sigma F_x = 0; \qquad 3\ \text{kN} - V_b = 0 \qquad V_b = 3\ \text{kN}$$



(c)



(d)

Average Shear Stress.

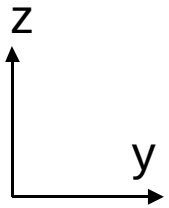
$$(\tau_a)_{\text{avg}} = \frac{V_a}{A_a} = \frac{3(10^3)\ \text{N}}{(0.1\ \text{m})(0.15\ \text{m})} = 200\ \text{kPa} \qquad \textit{Ans.}$$

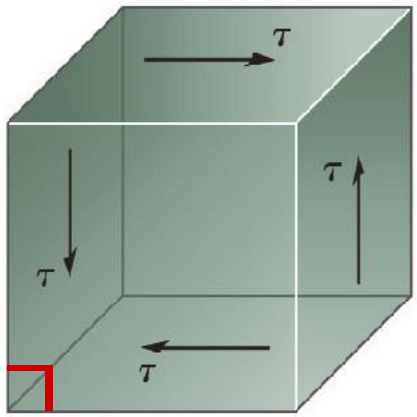
$$(\tau_b)_{\text{avg}} = \frac{V_b}{A_b} = \frac{3(10^3)\ \text{N}}{(0.125\ \text{m})(0.15\ \text{m})} = 160\ \text{kPa} \qquad \textit{Ans.}$$

The state of stress on elements located on sections *a*–*a* and *b*–*b* is shown in Figs. 1–23*c* and 1–23*d*, respectively.

5. Deformation en cisaillement

5.1 Définition:





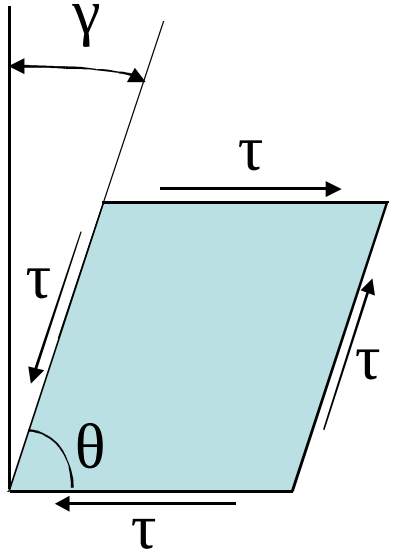
$\frac{\Pi}{2}$

Soit un cube soumis à une contrainte de cisaillement. Sous l'action de τ, l'angle originellement droit du cube change pour un angle θ.

La déformation de cisaillement γ est définie comme étant le changement d'angle d'un angle initialement droit.

$$\gamma = \frac{\Pi}{2} - \theta$$

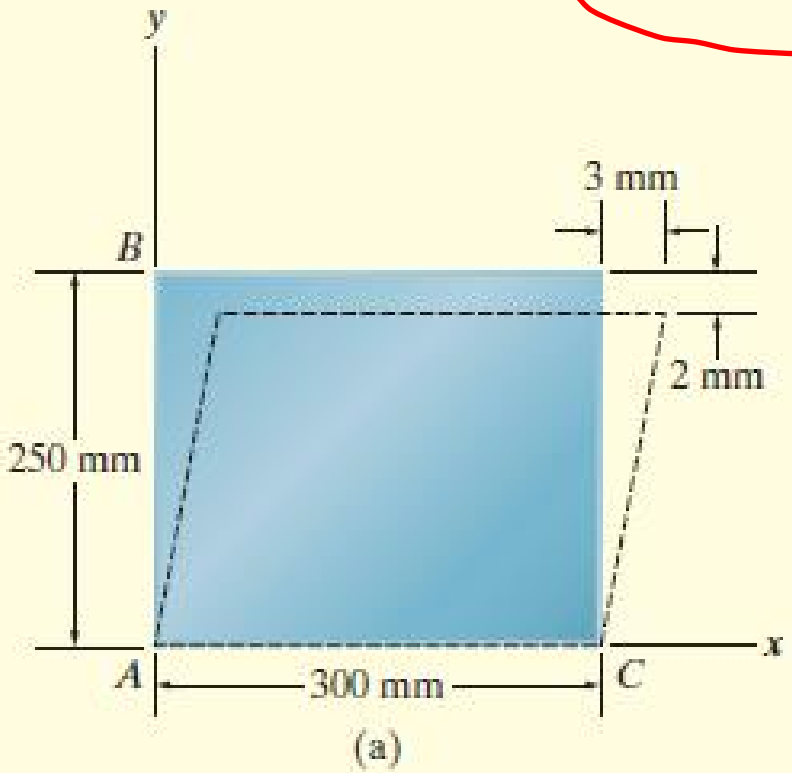
Unité: Radians



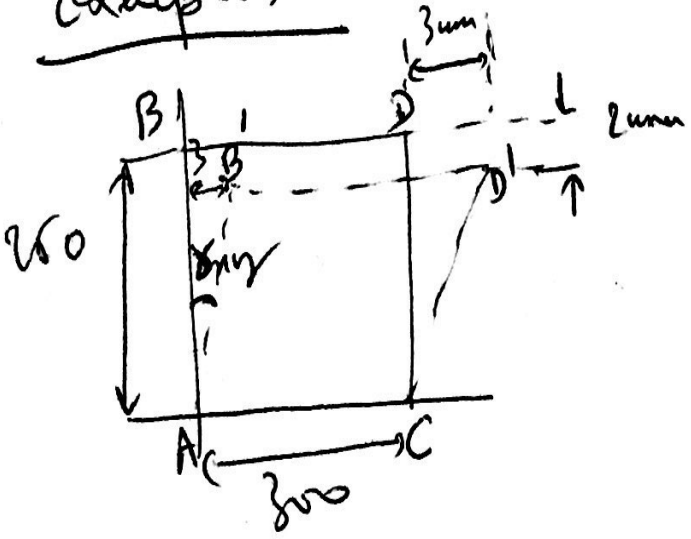
5.4 Exemple 7

Due to a loading, the plate is deformed into the dashed shape shown in Fig. 2–6*a*. Determine (a) the average normal strain along the side *AB*, and (b) the average shear strain in the plate at *A* relative to the *x* and *y* axes.

Strain = D®formation
Average = moyennne



Example 7:



$$AB' = \sqrt{(AB - \cancel{AB})^2 + BC^2}$$

$$= \sqrt{(250-2)^2 + 3^2}$$

$$AB' = 248,018\,mm = CD'$$

$$\varepsilon_{AB} = \frac{AB' - AB}{AB} = -0,00793.$$

↳ contraction

$$\tan\gamma = \frac{3}{250-2} \Rightarrow \gamma = 0,68^\circ = 0,012\,Rad$$

$$= \frac{0,68 \times \pi}{180^\circ}\,rad$$

$3,14\,rad \rightarrow 180^\circ$

1°

(4)

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6. Contrainte admissible

6.1 Facteur de sécurité

$$F.S. = \frac{\text{Charge à la rupture}}{\text{Charge admissible}}$$

6.2 Contrainte admissible (σ_{adm})

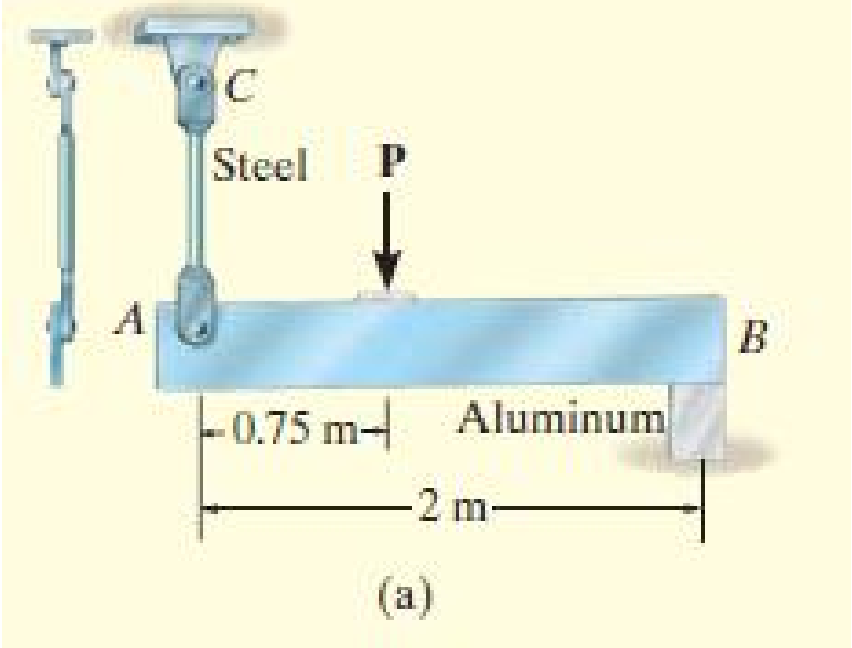
Si la relation entre la charge appliquée et les contraintes développées est linéaire alors:

$$F.S. = \frac{\sigma_{rup}}{\sigma_{adm}}$$

$$F.S. = \frac{\tau_{rup}}{\tau_{adm}}$$

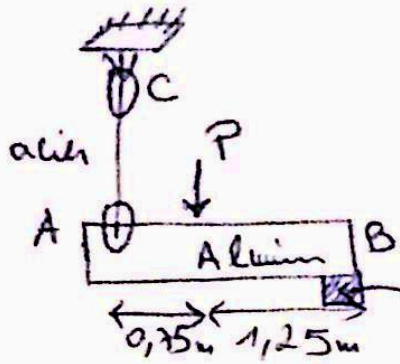
6.3 Exemple 8

The rigid bar AB shown in Fig. 1–29a is supported by a steel rod AC having a diameter of 20 mm and an aluminum block having a cross-sectional area of 1800 mm^2. The 18-mm-diameter pins at A and C are subjected to *single shear*. If the failure stress for the steel and aluminum is $(\sigma_{st})_{fail} = 680$ MPa and $(\sigma_{al})_{fail} = 70$ MPa, respectively, and the failure shear stress for each pin is $\tau_{fail} = 900$ MPa, determine the largest load P that can be applied to the bar. Apply a factor of safety of F.S. = 2.



(a)

Exemple 8



AB : Rigide → ne se déforme pas

C : Cisaillement triple

A : Cisaillement triple

Tige AC : diamètre 20mm

Bloc Alumi : 1800mm²

Rivet A et C : diamètre 18mm



cisaillement triple

$(\sigma_{Acier})_{rup} = 680 MPa$; $(\sigma_{Al})_{rup} = 70 MPa$

$(\tau_{Rivet})_{rup} = 900 MPa$; $FS = 2$

Valeur maximale de P qui peut être appliquée, sans provoquer une rupture ?

Rép :

$$(\sigma_{ad})_{acier} = \frac{(\sigma_{rup})_{acier}}{F.S} = \frac{680}{2} = 340 MPa$$

$$(\sigma_{ad})_{Al} = \frac{(\sigma_{rup})_{Al}}{F.S} = \frac{70}{2} = 35 MPa$$

$$(\tau_{ad})_{Rivet} = \frac{(\tau_{rup})_{Rivet}}{F.S} = \frac{900}{2} = 450 MPa$$

(*) $\sum \vec{M}_A = \vec{0} \Rightarrow -0,75P + 2F_B = 0 \Rightarrow P_{ad} = \frac{2(F_B)_{ad}}{0,75}$

or $(F_B)_{ad} = (\sigma_{ad})_{Al} \times A_B = 35.10^6 Pa \times 1800.10^{-6} m^2$

$= 63000 N = 63 kN$

Donc $P_{ad} = \frac{2 \times 63 kN}{0,75} = \boxed{168 kN}$.

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(*) $\sum \vec{M}_B = \vec{0} \Rightarrow 1{,}25P - 2F_A = 0$

$\Rightarrow P_{ad} = \dfrac{2(F_A)_{ad}}{1{,}25}$ or $(F_A)_{ad} = (\sigma_{ad})_{acier} \times A_{AC}$

$= (10^6 \times 340) \times (10.10^{-3})^2 \pi$

$= 106{,}814\ kN$

Donc $P_{ad} = \dfrac{2 \times 106{,}814}{1{,}25} = \boxed{170{,}902\ kN}$

(*) $(\tau_{Rivet})_{ad} = 450\ MPa \Rightarrow (V_{ad})_{Rivet} = 450.10^6 \times (9.10^{-3})^2 \pi$

$= 114511\ N$

$= 114{,}511\ kN$

$\sum \vec{M}_B = \vec{0} \Rightarrow 1{,}25P - 2V_A = 0$

$\Rightarrow P_{ad} = \dfrac{2(V_A)_{ad}}{1{,}25} = \dfrac{2 \times 114{,}511\ kN}{1{,}25}$

$= \boxed{183{,}217\ kN}$

Nous avons trouvé 3 valeurs de P_{ad} :

- 168 kN ou maximum pour ne pas casser la tige AC
- 170,902 kN " " " le bloc en Alumin
- 183,217 kN " " " les Rivets A et C

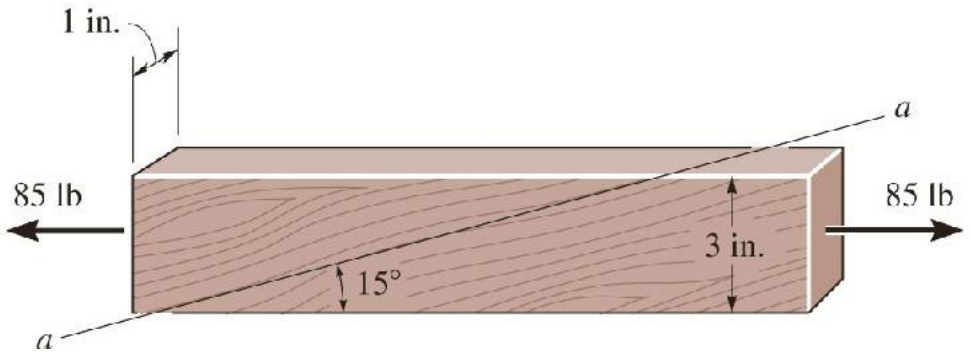
$\Rightarrow$ on prend le P_{ad} minimal $\Rightarrow P_{ad} = 168\ kN$

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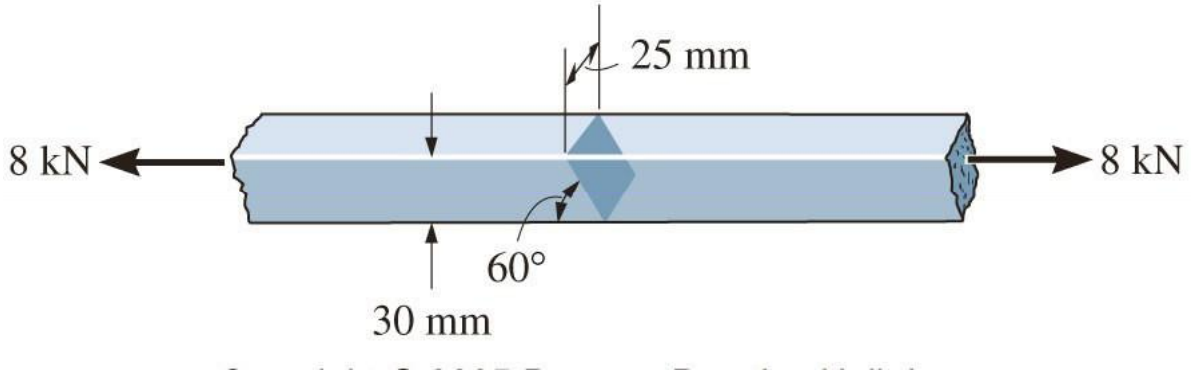
7. Contrainte sur des plans inclines

- Il est necessaire pour certaines applications de determiner les contraintes sur des plans inclines.
- Exemples:

Pièce de bois



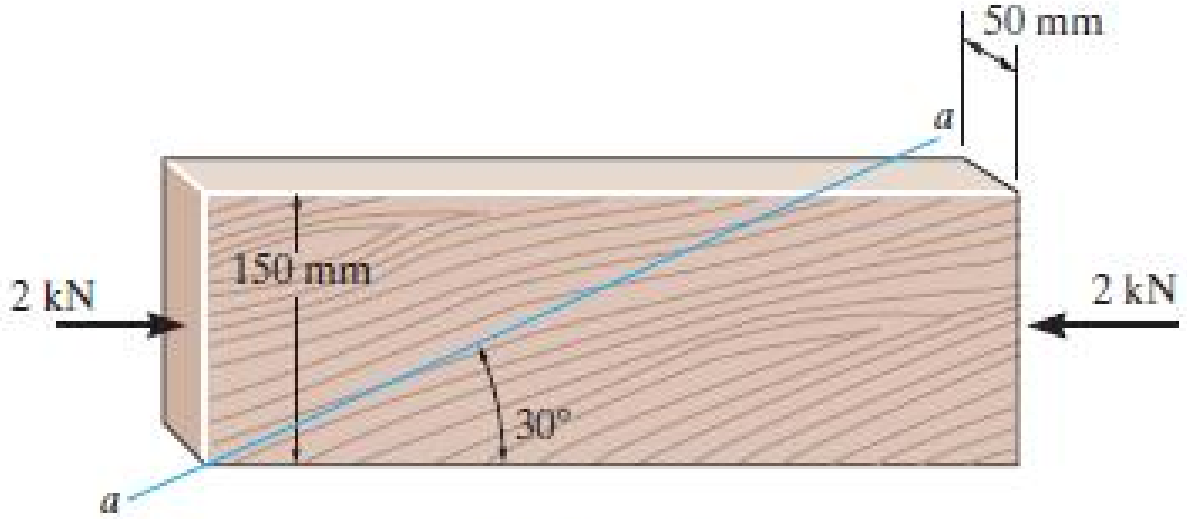
Pièce soudée



7. Contrainte sur des plans inclines

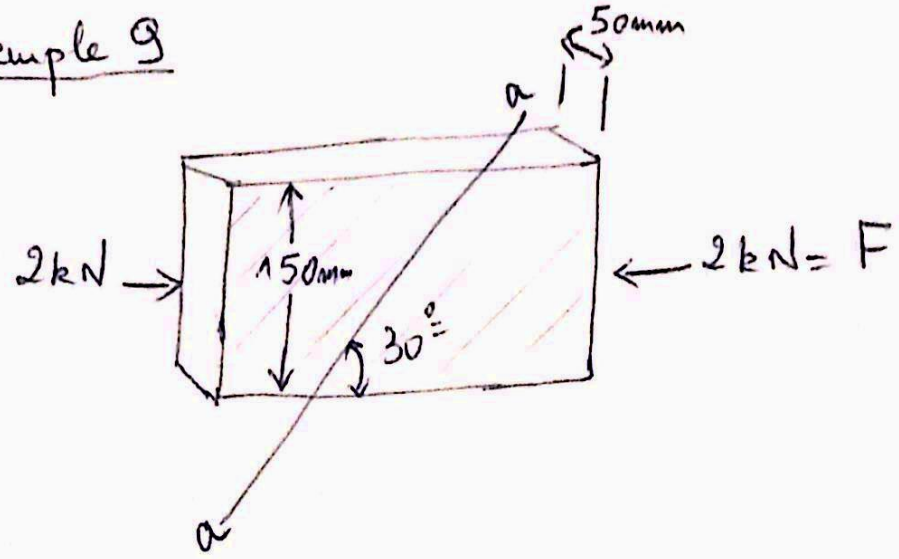
- **Exemple: 9**

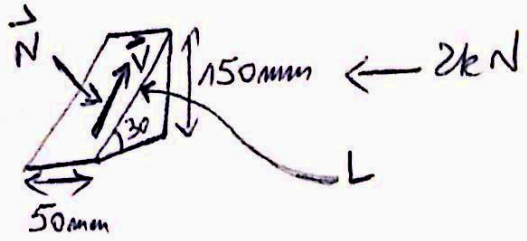
1–50. The block is subjected to a compressive force of 2 kN. Determine the average normal and average shear stress developed in the wood fibers that are oriented along section *a–a* at 30° with the axis of the block.



Prob. 1–50

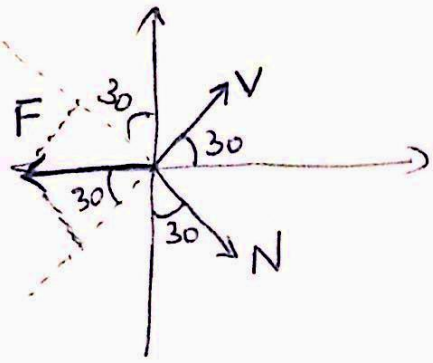
Exemple 9





$$\sin 30 = \frac{150}{L} \Rightarrow L = \frac{150}{\sin 30} = 300mm$$

$$\text{Donc } A = 300 \times 50 = 15000mm^2 = 0,015m^2$$



$$\sum \vec{F} = \vec{0} \Rightarrow \vec{N} + \vec{V} + \vec{F} = \vec{0}$$

$$F\cos 30 - V = 0$$
$$N - F\sin 30 = 0$$

$$\Rightarrow V = F\cos 30 \text{ et } N = F\sin 30$$

$$\sigma_{aa} = \frac{N}{A} = \frac{F\sin 30}{A} = \frac{2.10^3 \times 0,5}{0,015} = 66,67kPa$$

$$\tau = \frac{V}{A} = \frac{F\cos 30}{A} = \frac{2.10^3 \times 0,86}{0,015} = 115,47kPa$$

Remarque : $30 = \alpha$; $B = 50mm$, $A = BL = Bh/\sin\alpha$; $V = F\cos\alpha$, $N = F\sin\alpha$

$\sigma = N/A = F\sin^2\alpha/Bh$; $\tau = V/A = F\cos\alpha\sin\alpha/Bh \Rightarrow \tau_{max} = \frac{1}{4}$

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8. Conclusion

Jusqu'à maintenant, on a vu les notions suivantes:

- Sollicitations
- Réactions aux appuis
- Équations d'équilibre
- Diagramme de corps libre
- Forces internes
- **Contraintes (σ, τ)**
- **Déformations (ε, γ)**
- **Déplacements, changement de forme**

Vous devez bien comprendre ces notions avant de continuer.

